Kerala Polytechnic Revision 2021 • Premium Lesson File

Applied Physics Lab

Diploma in Engineering and Technology • Semester 1 & 2 • Basic Science

Course Code

1006

Base course code: 1006

Credits: 2 • L:0, T:0, P:2

Periods/Semester: 30 + 30

Subject Type

Physics Lab

This file adapts the lesson structure according to the subject type, so the learning material behaves like a textbook, revision guide, workbook and answer guide.

Programme

Diploma in Engineering and Technology

Designed for Kerala Polytechnic students with simple explanations, exam-friendly formats and practical support notes.

Exam / Evaluation

Evaluation Style

Minimum 12 experiments; CIA in both semesters; ESE at end of second semester.

Student-friendly subject summary

Applied Physics Lab builds the foundation needed for the semester by connecting syllabus outcomes with real study practice. Read the overview first, then open one module at a time, revise key points, and finally practise the expected questions with the master answer key.

Malayalam tip: □□□ module-□□ open □□□□□□ □□□□□□□□. □□□□□□ □□□□□□□□□

□□□□□□□□□□□□□□□□ focus □□□□□□□□□□□□□□.

Prerequisite Knowledge

- Basic knowledge in Physics from secondary school.

Course Objectives

- Supplement physics theory through first-hand manipulation of apparatus.
- Develop scientific temper for engineering and technology problems.
- Build confidence in handling equipment and measurement skills.

How to study this subject

1. Read the module outcome before studying the topic.
2. Write definitions, formulas/key points and procedures in a separate revision page.
3. Practise expected questions after completing each module.
4. Use the Master Answer Key to improve answer order and presentation.

Course Outcomes

CO	Outcome	Hours	Level
CO1	Select measuring tools and make accurate measurements.	12	Applying
CO2	Apply mechanics and properties of matter through experiments.	23	Applying
CO3	Use lens, prism and glass slab to realise ray optics laws.	6	Applying
CO4	Use V-I characteristics to determine resistance of materials.	15	Applying

Module Map

Module	Title	Hours	Linked outcome
Module 1	Measurement Tools: Vernier, Screw Gauge, Spherometer and Thermometer	12	Select measuring tools and make accurate measurements.
Module 2	Mechanics and Properties of Matter Experiments	23	Apply mechanics and properties of matter through experiments.
Module 3	Ray Optics Experiments	6	Use lens, prism and glass slab to realise ray optics laws.
Module 4	Electrical and Semiconductor Experiments	15	Use V-I characteristics to determine resistance of materials.

Quick navigation cards

Mini Formula Bank

Mini Formula Bank for Applied Physics Lab

Use this as the last-minute memory page before exams and practical sessions.

Least count

```
Smallest measurable value of an instrument
```

Where used: Measurement experiments

Common mistake: Write least count before observations.

Hooke's law

```
F = kx
```

Where used: Spring experiment

Common mistake: Valid only within elastic limit.

Pendulum relation

```
T = 2π√(l/g)
```

Where used: Simple pendulum

Common mistake: Length must be from point of suspension to bob centre.

Ohm's law

```
V = IR
```

Where used: V-I characteristics

Common mistake: Temperature should remain nearly constant.

Series resistance

```
R = R₁ + R₂ + ...
```

Where used: Resistance law

Common mistake: Series increases total resistance.

Parallel resistance

```
1/R = 1/R₁ + 1/R₂ + ...
```

Where used: Resistance law

Common mistake: Equivalent is less than smallest branch resistance.

Lens power

```
P = 1/f(m)
```

Where used: Optics

Common mistake: Focal length must be in metre.

Module 1 • 12 hours

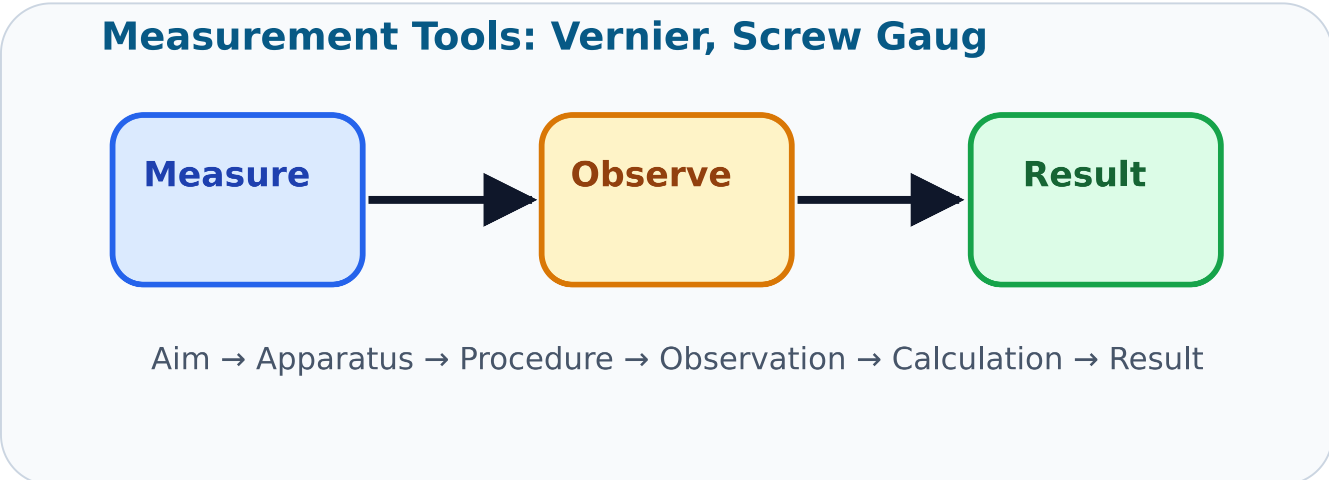
Measurement Tools: Vernier, Screw Gauge, Spherometer and Thermometer

Learning outcome: Select measuring tools and make accurate measurements.

Lab record-□ least count, zero error, units □□□□□□ skip □□□□□□□□□.

Simple introduction

This module should be studied as a connected set of ideas, not as isolated points. First understand the purpose, then learn the correct terms, then practise writing or performing the answer in exam/lab format.



Topic-wise explanation

Vernier calipers volume measurement

Vernier calipers volume measurement is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

Screw gauge diameter/thickness

Screw gauge diameter/thickness is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

Spherometer radius of curvature

Spherometer radius of curvature is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

Mercury thermometer

Mercury thermometer is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

Least count

Least count is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

Zero error

Zero error is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

Unit conversion

Unit conversion is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

Observation table

Observation table is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

Important terms

Term	Exam-ready meaning
Vernier calipers volume measurement	Vernier calipers volume measurement is a core point in this module. In exam answers, connect it with Measurement Tools: Vernier, Screw Gauge, Spherometer and Thermometer and give one clear example.
Screw gauge diameter/thickness	Screw gauge diameter/thickness is a core point in this module. In exam answers, connect it with Measurement Tools: Vernier, Screw Gauge, Spherometer and Thermometer and give one clear example.
Spherometer radius of curvature	Spherometer radius of curvature is a core point in this module. In exam answers, connect it with Measurement Tools: Vernier, Screw Gauge, Spherometer and Thermometer and give one clear example.
Mercury thermometer	Mercury thermometer is a core point in this module. In exam answers, connect it with Measurement Tools: Vernier, Screw Gauge, Spherometer and Thermometer and give one clear example.
Least count	Least count is a core point in this module. In exam answers, connect it with Measurement Tools: Vernier, Screw Gauge, Spherometer and Thermometer and give one clear example.

Practical / real-life examples

Example

Volume of cylinder = $\pi r^2 h$ using vernier readings.

Example

Correct reading = observed reading ± zero correction.

Exam writing tip: Start with the exact term, then add explanation, example and conclusion.

Common mistakes

- Writing answers without the required order or headings.
- Skipping units, labels, procedure steps or diagrams where needed.
- Memorising without understanding the application.
- Not matching the answer to the question verb: explain, compare, draw, calculate or list.

Module checklist

Outcome read | **Definitions ready** | **Examples practised** | **Diagram/table revised** | **Questions attempted** | **Mistakes checked**

Expected Questions

1. **M1-Q1** Explain Vernier calipers volume measurement with its importance in Measurement Tools: Vernier, Screw Gauge, Spherometer and Thermometer.
2. **M1-Q2** Write short notes on Screw gauge diameter/thickness.
3. **M1-Q3** Prepare an exam-ready answer for Spherometer radius of curvature.
4. **M1-Q4** Compare or connect two key ideas from Measurement Tools: Vernier, Screw Gauge, Spherometer and Thermometer.

5. **M1-Q5** Draw/describe a neat labelled diagram, table or format related to Measurement Tools: Vernier, Screw Gauge, Spherometer and Thermometer.

6. **M1-Q6** List common mistakes/precautions while studying or performing Measurement Tools: Vernier, Screw Gauge, Spherometer and Thermometer.

Module 2 • 23 hours

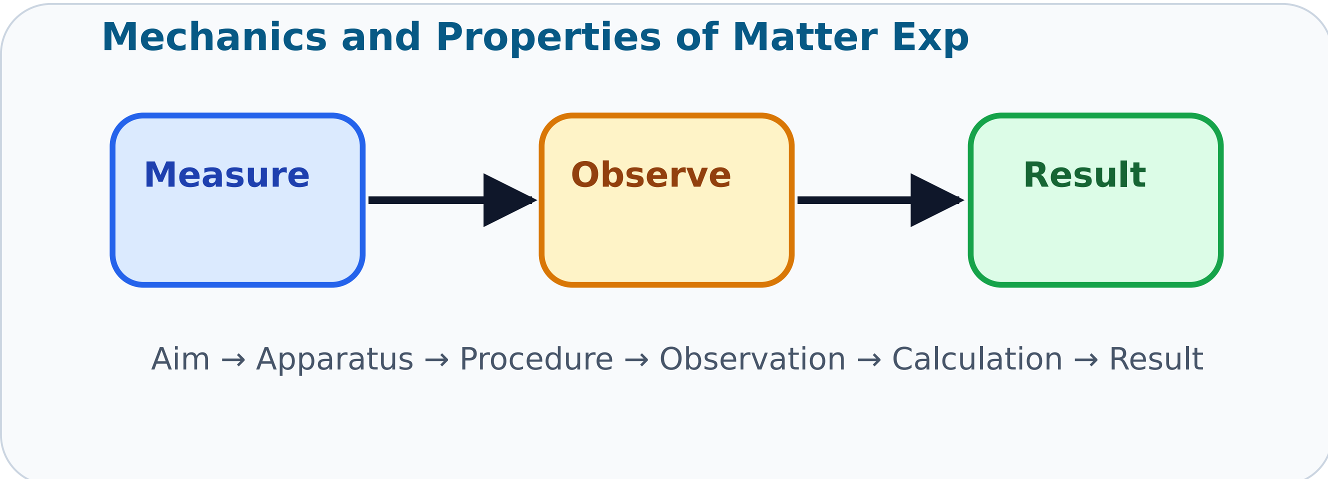
Mechanics and Properties of Matter Experiments

Learning outcome: Apply mechanics and properties of matter through experiments.

Repeated readings □□□□□□□ mean value □□□□□□□□□□□ result reliable □□□□.

Simple introduction

This module should be studied as a connected set of ideas, not as isolated points. First understand the purpose, then learn the correct terms, then practise writing or performing the answer in exam/lab format.



Topic-wise explanation

Hooke's law and spring constant

Hooke's law and spring constant is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

Stoke's law viscosity

Stoke's law viscosity is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

Parallelogram law of forces

Parallelogram law of forces is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

Moment bar

Moment bar is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

U-tube relative density

U-tube relative density is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

Flywheel moment of inertia

Flywheel moment of inertia is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

Simple pendulum g

Simple pendulum g is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

Resonance column velocity of sound

Resonance column velocity of sound is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

Cantilever oscillation

Cantilever oscillation is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

Important terms

Term	Exam-ready meaning
Hooke's law and spring constant	Hooke's law and spring constant is a core point in this module. In exam answers, connect it with Mechanics and Properties of Matter Experiments and give one clear example.
Stoke's law viscosity	Stoke's law viscosity is a core point in this module. In exam answers, connect it with Mechanics and Properties of Matter Experiments and give one clear example.
Parallelogram law of forces	Parallelogram law of forces is a core point in this module. In exam answers, connect it with Mechanics and Properties of Matter Experiments and give one clear example.
Moment bar	Moment bar is a core point in this module. In exam answers, connect it with Mechanics and Properties of Matter Experiments and give one clear example.
U-tube relative density	U-tube relative density is a core point in this module. In exam answers, connect it with Mechanics and Properties of Matter Experiments and give one clear example.

Practical / real-life examples

Example

Hooke's law graph: load vs extension should be a straight line within elastic limit.

Example

Simple pendulum: measure time for many oscillations to reduce error.

Exam writing tip: Start with the exact term, then add explanation, example and conclusion.

Common mistakes

- Writing answers without the required order or headings.
- Skipping units, labels, procedure steps or diagrams where needed.
- Memorising without understanding the application.
- Not matching the answer to the question verb: explain, compare, draw, calculate or list.

Module checklist

Outcome read | Definitions ready | Examples practised

Diagram/table revised | Questions attempted | Mistakes checked

Expected Questions

1. **M2-Q1** Explain Hooke's law and spring constant with its importance in Mechanics and Properties of Matter Experiments.
2. **M2-Q2** Write short notes on Stoke's law viscosity.
3. **M2-Q3** Prepare an exam-ready answer for Parallelogram law of forces.
4. **M2-Q4** Compare or connect two key ideas from Mechanics and Properties of Matter Experiments.
5. **M2-Q5** Draw/describe a neat labelled diagram, table or format related to Mechanics and Properties of Matter Experiments.
6. **M2-Q6** List common mistakes/precautions while studying or performing Mechanics and Properties of Matter Experiments.

Module 3 • 6 hours

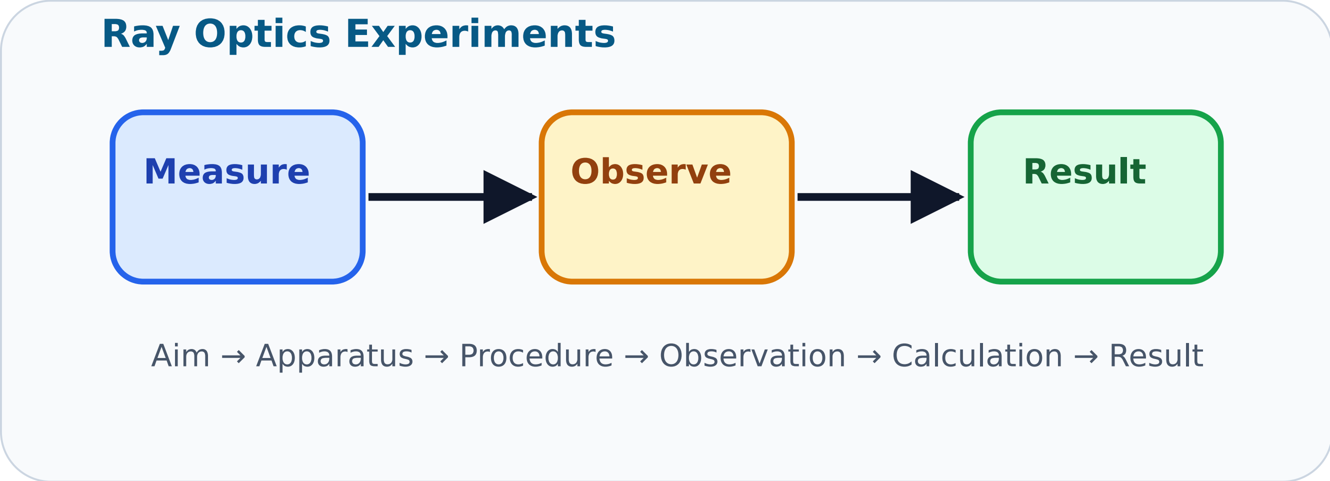
Ray Optics Experiments

Learning outcome: Use lens, prism and glass slab to realise ray optics laws.

Optics-□ pin alignment/parallax error marks loss □□□□ □□□□□□□□□.

Simple introduction

This module should be studied as a connected set of ideas, not as isolated points. First understand the purpose, then learn the correct terms, then practise writing or performing the answer in exam/lab format.



Topic-wise explanation

Convex lens focal length by u-v method

Convex lens focal length by u-v method is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

Power of lens

Power of lens is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

Snell's law

Snell's law is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

Glass slab/prism refraction

Glass slab/prism refraction is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

Object and image distance

Object and image distance is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

Ray diagram

Ray diagram is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

Refractive index

Refractive index is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

Important terms

Term	Exam-ready meaning
Convex lens focal length by u-v method	Convex lens focal length by u-v method is a core point in this module. In exam answers, connect it with Ray Optics Experiments and give one clear example.
Power of lens	Power of lens is a core point in this module. In exam answers, connect it with Ray Optics Experiments and give one clear example.
Snell's law	Snell's law is a core point in this module. In exam answers, connect it with Ray Optics Experiments and give one clear example.
Glass slab/prism refraction	Glass slab/prism refraction is a core point in this module. In exam answers, connect it with Ray Optics Experiments and give one clear example.
Object and image distance	Object and image distance is a core point in this module. In exam answers, connect it with Ray Optics Experiments and give one clear example.

Practical / real-life examples

Example

Lens formula: 1/f = 1/v - 1/u (sign convention as taught).

Example

Power P = 1/f in metre, unit dioptre.

Exam writing tip: Start with the exact term, then add explanation, example and conclusion.

Common mistakes

- Writing answers without the required order or headings.
- Skipping units, labels, procedure steps or diagrams where needed.
- Memorising without understanding the application.
- Not matching the answer to the question verb: explain, compare, draw, calculate or list.

Module checklist

- Outcome read
- Definitions ready
- Examples practised
- Diagram/table revised
- Questions attempted
- Mistakes checked

Expected Questions

1. **M3-Q1** Explain Convex lens focal length by u-v method with its importance in Ray Optics Experiments.
2. **M3-Q2** Write short notes on Power of lens.
3. **M3-Q3** Prepare an exam-ready answer for Snell's law.
4. **M3-Q4** Compare or connect two key ideas from Ray Optics Experiments.
5. **M3-Q5** Draw/describe a neat labelled diagram, table or format related to Ray Optics Experiments.
6. **M3-Q6** List common mistakes/precautions while studying or performing Ray Optics Experiments.

Module 4 • 15 hours

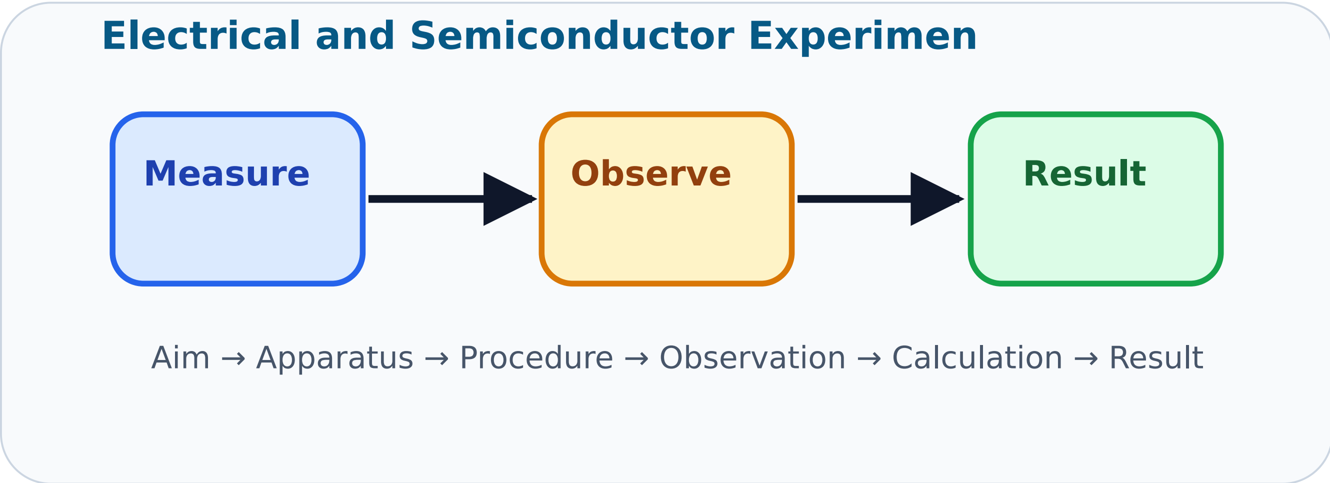
Electrical and Semiconductor Experiments

Learning outcome: Use V-I characteristics to determine resistance of materials.

Circuit ON □□□□□□□□□□□□□ □□□□□□ connection faculty/technician□□ □□□□□□□□ verify □□□□□□□.

Simple introduction

This module should be studied as a connected set of ideas, not as isolated points. First understand the purpose, then learn the correct terms, then practise writing or performing the answer in exam/lab format.



Topic-wise explanation

Ohm's law V-I graph

Ohm's law V-I graph is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

Resistance in series and parallel

Resistance in series and parallel is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

Semiconductor diode V-I characteristics

Semiconductor diode V-I characteristics is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

Knee voltage

Knee voltage is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

Potentiometer internal resistance/emf comparison

Potentiometer internal resistance/emf comparison is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

Meter bridge specific resistance

Meter bridge specific resistance is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

Circuit connections

Circuit connections is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

Graph plotting

Graph plotting is part of the official syllabus for this module. Study it by writing the definition/purpose, the working idea or procedure, one application, and one common error to avoid. In exam answers, keep the order logical and use labelled points.

Important terms

Term	Exam-ready meaning
Ohm's law V-I graph	Ohm's law V-I graph is a core point in this module. In exam answers, connect it with Electrical and Semiconductor Experiments and give one clear example.
Resistance in series and parallel	Resistance in series and parallel is a core point in this module. In exam answers, connect it with Electrical and Semiconductor Experiments and give one clear example.
Semiconductor diode V-I characteristics	Semiconductor diode V-I characteristics is a core point in this module. In exam answers, connect it with Electrical and Semiconductor Experiments and give one clear example.
Knee voltage	Knee voltage is a core point in this module. In exam answers, connect it with Electrical and Semiconductor Experiments and give one clear example.
Potentiometer internal resistance/emf comparison	Potentiometer internal resistance/emf comparison is a core point in this module. In exam answers, connect it with Electrical and Semiconductor Experiments and give one clear example.

Practical / real-life examples

Example

Ohm's law: slope of V-I graph gives resistance.

Example

Meter bridge balance point is used to calculate unknown resistance.

Exam writing tip: Start with the exact term, then add explanation, example and conclusion.

Common mistakes

- Writing answers without the required order or headings.
- Skipping units, labels, procedure steps or diagrams where needed.
- Memorising without understanding the application.
- Not matching the answer to the question verb: explain, compare, draw, calculate or list.

Module checklist

Outcome read **Definitions ready** **Examples practised**

Diagram/table revised **Questions attempted** **Mistakes checked**

Expected Questions

1. **M4-Q1** Explain Ohm's law V-I graph with its importance in Electrical and Semiconductor Experiments.
2. **M4-Q2** Write short notes on Resistance in series and parallel.
3. **M4-Q3** Prepare an exam-ready answer for Semiconductor diode V-I characteristics.
4. **M4-Q4** Compare or connect two key ideas from Electrical and Semiconductor Experiments.
5. **M4-Q5** Draw/describe a neat labelled diagram, table or format related to Electrical and Semiconductor Experiments.
6. **M4-Q6** List common mistakes/precautions while studying or performing Electrical and Semiconductor Experiments.

Quick Revision

One-page revision summary

Use this section on the previous day of exam or practical assessment.

Module-wise must-know points

- **Module 1:** Measurement Tools: Vernier, Screw Gauge, Spherometer and Thermometer — revise Vernier calipers volume measurement, Screw gauge diameter/thickness, Spherometer radius of curvature, Mercury thermometer.

- **Module 2:** Mechanics and Properties of Matter Experiments — revise Hooke's law and spring constant, Stoke's law viscosity, Parallelogram law of forces, Moment bar.
- **Module 3:** Ray Optics Experiments — revise Convex lens focal length by u-v method, Power of lens, Snell's law, Glass slab/prism refraction.
- **Module 4:** Electrical and Semiconductor Experiments — revise Ohm's law V-I graph, Resistance in series and parallel, Semiconductor diode V-I characteristics, Knee voltage.

Important definitions / formulas / key points

- **Least count:** Smallest measurable value of an instrument — Measurement experiments
- **Hooke's law:** $F = kx$ — Spring experiment
- **Pendulum relation:** $T = 2\pi\sqrt{(l/g)}$ — Simple pendulum
- **Ohm's law:** $V = IR$ — V-I characteristics
- **Series resistance:** $R = R_1 + R_2 + \ldots$ — Resistance law
- **Parallel resistance:** $1/R = 1/R_1 + 1/R_2 + \ldots$ — Resistance law
- **Lens power:** $P = 1/f(m)$ — Optics

Common exam mistakes

- Writing general answers instead of syllabus-topic answers.
- Forgetting diagrams, tables, units, procedures or examples.
- Skipping final conclusion/result in long answers.
- Not numbering answers neatly.

Last-minute checklist

Course outcomes | **Module map** | **Key bank** | **Expected questions**

Answer format | **Diagrams/tables** | **Final revision**

Diagram / table list

Prepare neat labelled diagrams, flowcharts, observation tables, comparison tables or answer formats wherever the module asks for drawing, procedure, calculation or communication structure.

Master Answer Key

Answers matching every expected question

Each answer number exactly follows the expected question number.

M1-Q1 — Explain Vernier calipers volume measurement with its importance in Measurement Tools: Vernier, Screw Gauge, Spherometer and Thermometer.

Answer format: Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end with precautions. For this module, cover: Vernier calipers volume measurement, Screw gauge diameter/thickness, Spherometer radius of curvature, Mercury thermometer, Least count.

Module link: Measurement Tools: Vernier, Screw Gauge, Spherometer and Thermometer

M1-Q2 — Write short notes on Screw gauge diameter/thickness.

Answer format: Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end with precautions. For this module, cover: Vernier calipers volume measurement, Screw gauge diameter/thickness, Spherometer radius of curvature, Mercury thermometer, Least count.

Module link: Measurement Tools: Vernier, Screw Gauge, Spherometer and Thermometer

M1-Q3 — Prepare an exam-ready answer for Spherometer radius of curvature.

Answer format: Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end with precautions. For this module, cover: Vernier calipers volume measurement, Screw gauge diameter/thickness, Spherometer radius of curvature, Mercury thermometer, Least count.

Module link: Measurement Tools: Vernier, Screw Gauge, Spherometer and Thermometer

M1-Q4 — Compare or connect two key ideas from Measurement Tools: Vernier, Screw Gauge, Spherometer and Thermometer.

Answer format: Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end with precautions. For this module, cover: Vernier calipers volume measurement, Screw gauge diameter/thickness, Spherometer radius of curvature, Mercury thermometer, Least count.

Module link: Measurement Tools: Vernier, Screw Gauge, Spherometer and Thermometer

M1-Q5 — Draw/describe a neat labelled diagram, table or format related to Measurement Tools: Vernier, Screw Gauge, Spherometer and Thermometer.

Answer format: Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end with precautions. For this module, cover: Vernier calipers volume measurement, Screw gauge diameter/thickness, Spherometer radius of curvature, Mercury thermometer, Least count.

Module link: Measurement Tools: Vernier, Screw Gauge, Spherometer and Thermometer

M1-Q6 — List common mistakes/precautions while studying or performing Measurement Tools: Vernier, Screw Gauge, Spherometer and Thermometer.

Answer format: Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end with precautions. For this module, cover: Vernier calipers volume measurement, Screw gauge diameter/thickness, Spherometer radius of curvature, Mercury thermometer, Least count.

Module link: Measurement Tools: Vernier, Screw Gauge, Spherometer and Thermometer

M2-Q1 — Explain Hooke's law and spring constant with its importance in Mechanics and Properties of Matter Experiments.

Answer format: Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end with precautions. For this module, cover: Hooke's law and spring constant, Stoke's law viscosity, Parallelogram law of forces, Moment bar, U-tube relative density.

Module link: Mechanics and Properties of Matter Experiments

M2-Q2 — Write short notes on Stoke's law viscosity.

Answer format: Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end with precautions. For this module, cover: Hooke's law and spring constant, Stoke's law viscosity, Parallelogram law of forces, Moment bar, U-tube relative density.

Module link: Mechanics and Properties of Matter Experiments

M2-Q3 — Prepare an exam-ready answer for Parallelogram law of forces.

Answer format: Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end with precautions. For this module, cover: Hooke's law and spring constant, Stoke's law viscosity, Parallelogram law of forces, Moment bar, U-tube relative density.

Module link: Mechanics and Properties of Matter Experiments

M2-Q4 — Compare or connect two key ideas from Mechanics and Properties of Matter Experiments.

Answer format: Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end with precautions. For this module, cover: Hooke's law and spring constant, Stoke's law viscosity, Parallelogram law of forces, Moment bar, U-tube relative density.

Module link: Mechanics and Properties of Matter Experiments

M2-Q5 — Draw/describe a neat labelled diagram, table or format related to Mechanics and Properties of Matter Experiments.

Answer format: Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end with precautions. For this module, cover: Hooke's law and spring constant, Stoke's law viscosity, Parallelogram law of forces, Moment bar, U-tube relative density.

Module link: Mechanics and Properties of Matter Experiments

M2-Q6 — List common mistakes/precautions while studying or performing Mechanics and Properties of Matter Experiments.

Answer format: Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end with precautions. For this module, cover: Hooke's law and spring constant, Stoke's law viscosity, Parallelogram law of forces, Moment bar, U-tube relative density.

Module link: Mechanics and Properties of Matter Experiments

M3-Q1 — Explain Convex lens focal length by u-v method with its importance in Ray Optics Experiments.

Answer format: Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end with precautions. For this module, cover: Convex lens focal length by u-v method, Power of lens, Snell's law, Glass slab/prism refraction, Object and image distance.

Module link: Ray Optics Experiments

M3-Q2 — Write short notes on Power of lens.

Answer format: Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end with precautions. For this module, cover: Convex lens focal length by u-v method, Power of lens, Snell's law, Glass slab/prism refraction, Object and image distance.

Module link: Ray Optics Experiments

M3-Q3 — Prepare an exam-ready answer for Snell's law.

Answer format: Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end with precautions. For this module, cover: Convex lens focal length by u-v method, Power of lens, Snell's law, Glass slab/prism refraction, Object and image distance.

Module link: Ray Optics Experiments

M3-Q4 — Compare or connect two key ideas from Ray Optics Experiments.

Answer format: Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end with precautions. For this module, cover: Convex lens focal length by u-v method, Power of lens, Snell's law, Glass slab/prism refraction, Object and image distance.

Module link: Ray Optics Experiments

M3-Q5 — Draw/describe a neat labelled diagram, table or format related to Ray Optics Experiments.

Answer format: Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end with precautions. For this module, cover: Convex lens focal length by u-v method, Power of lens, Snell's law, Glass slab/prism refraction, Object and image distance.

Module link: Ray Optics Experiments

M3-Q6 — List common mistakes/precautions while studying or performing Ray Optics Experiments.

Answer format: Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end with precautions. For this module, cover: Convex lens focal length by u-v method, Power of lens, Snell's law, Glass slab/prism refraction, Object and image distance.

Module link: Ray Optics Experiments

M4-Q1 — Explain Ohm's law V-I graph with its importance in Electrical and Semiconductor Experiments.

Answer format: Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end with precautions. For this module, cover: Ohm's law V-I graph, Resistance in series and parallel, Semiconductor diode V-I characteristics, Knee voltage, Potentiometer internal resistance/emf comparison.

Module link: Electrical and Semiconductor Experiments

M4-Q2 — Write short notes on Resistance in series and parallel.

Answer format: Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end with precautions. For this module, cover: Ohm's law V-I graph, Resistance in series and parallel, Semiconductor diode V-I characteristics, Knee voltage, Potentiometer internal resistance/emf comparison.

Module link: Electrical and Semiconductor Experiments

M4-Q3 — Prepare an exam-ready answer for Semiconductor diode V-I characteristics.

Answer format: Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end with precautions. For this module, cover: Ohm's law V-I graph, Resistance in series and parallel, Semiconductor diode V-I characteristics, Knee voltage, Potentiometer internal resistance/emf comparison.

Module link: Electrical and Semiconductor Experiments

M4-Q4 — Compare or connect two key ideas from Electrical and Semiconductor Experiments.

Answer format: Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end with precautions. For this module, cover: Ohm's law V-I graph, Resistance in series and parallel, Semiconductor diode V-I characteristics, Knee voltage, Potentiometer internal resistance/emf comparison.

Module link: Electrical and Semiconductor Experiments

M4-Q5 — Draw/describe a neat labelled diagram, table or format related to Electrical and Semiconductor Experiments.

Answer format: Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end with precautions. For this module, cover: Ohm's law V-I graph, Resistance in series and parallel, Semiconductor diode V-I characteristics, Knee voltage, Potentiometer internal resistance/emf comparison.

Module link: Electrical and Semiconductor Experiments

M4-Q6 — List common mistakes/precautions while studying or performing Electrical and Semiconductor Experiments.

Answer format: Start with the aim, then list required apparatus or software, write the procedure in correct order, record observations neatly, complete calculation/result where applicable, and end with precautions. For this module, cover: Ohm's law V-I graph, Resistance in series and parallel, Semiconductor diode V-I characteristics, Knee voltage, Potentiometer internal resistance/emf comparison.

Module link: Electrical and Semiconductor Experiments